CONTEXT/BACKGROUND

This is a research project aiming to address rental units' carbon footprint in the City of Vancouver. Our project overviews the current reality of renters' carbon footprint in Vancouver. This includes both causes of renters' carbon footprint, and intersectional challenges which may limit their ability to engage with carbon reduction programs. We also survey academic literature which explores reasons for high carbon footprint by renters - such as the split incentive problem - and how other jurisdictions have addressed this issue. Finally, our project outlines and evaluates several policy solutions which help reduce renters' carbon footprint using an equity lens.

This project received First Place at the Vancouver HUBBUB CityStudio Conference, which entailed a poster presentation and a half-day of engagement with diverse industry, government, and public stakeholders. Our project has already been completed through our coursework with the University of British Columbia's Masters Program in Political Science.

EXECUTIVE SUMMARY

In January 2019, Vancouver Council declared a climate emergency in recognition of urgent threats posed by climate change, proposing to strengthen Vancouver's efforts to cut carbon emissions. The current response framework, the Climate Emergency Action Plan (CEAP), targets emission reductions in ways that bring financial, health and economic benefits to Vancouver, setting the course to reduce the City's carbon emissions by 50% by 2030.[1] Climate justice and equity are also at the forefront of this Plan.[2]

Currently, burning natural gas and hot water in buildings (54% of the total in 2019) generates the highest carbon emission in the City of Vancouver, prompting a target to cut building carbon emissions in half from 2007 levels by 2030. To achieve this, the City of Vancouver must account for all types of residents, including households composed of renters. This memo defines a key policy problem: barriers to reducing renters' carbon emissions. To address this policy problem, we suggest five policy alternatives:

1. Rental Units Carbon Footprint Report Card;
2. Renters' Service Centre in the City of Vancouver;
3. Small Appliance Loan Program in the City of Vancouver;
4. Small Investment Grant for Renters; and
5. Revised Standards of Maintenance Bylaw to Carbon-Reducing Appliances.

Recommendation

We recommend that the City of Vancouver implement a "Rental Unit Carbon Emissions Report Card" to be shared with prospective renters, ideally including information concerning rental unit utility costs and carbon emissions. This solution could enable renters to make more informed choices when choosing rental units, based on associated utility costs and their

[1] "Council Report: Climate Emergency Action Plan," City of Vancouver, updated October 22, 2020, https://council.vancouver.ca/20201103/documents/p1.pdf.
[2] "Council Report: Climate Emergency Action Plan."

carbon outputs, while potentially encouraging landlords to invest in energy-efficient systems to compete in a more transparent rental market.

1. THE POLICY PROBLEM

1.1 Split Incentives for Landlords and Renters

Renters' carbon footprints are directly related to market failures of energy consumption. Split incentives between landlords and renters lead to inefficient energy consumption, driven by asymmetrical information and inattentiveness. As British Columbia shifts away from fossil fuel consumption to renewable energy sources,[3] the price of fossil fuels is likely to increase in the foreseeable future.[4] All residents in Vancouver must be able to engage in zero carbon energy to lower Vancouver's GHG emissions, and to ensure renters are not disproportionately impacted by the predicted rising cost of fossil fuels. Although our focus is on increasing renters' agency in reducing their carbon footprints, that does not preclude complementary measures aimed at landlords who own the buildings in question.

1.1.1 Energy Efficiency and Split Incentives

Issues of sharing utility costs between landlords and renters create a 'split incentive issue.' Landlords and renters make decisions on energy consumption within the household; however, only a single actor pays for the cost of their energy-consumption choices through the utility bill.[5] It is useful for us to consider three scenarios: one where renters pay the utility bill, one where the landlord pays, and one where the costs are split.

When utilities are paid by renters, landlords are significantly less likely to invest in energy-efficient appliances, as they are not faced with the utility costs.[6] This can lead to higher utility bills for renters due to inefficient technology, and higher carbon footprints. When utilities are included in the rent, renters are less likely to change their energy

[3] Canada, British Columbia, Ministry of Environment and Climate Change Strategy, *Clean B.C. Roadmap to 2030*, [Victoria], 2021, https://www2.gov.bc.ca/assets/gov/environment/climate-change/action/cleanbc/cleanbc_roadmap_2030.pdf
[4] The Greenlining Institute, *Equitable Building Electrification: A Framework for Powering Resilient Communities*. (Oakland: The Greenlining Institute, 2019). http://greenlining.org/wp-content/uploads/2019/09/Greenlining_EquitableElectrificationReport_2019_WEB.pdf.
[5] Jesse Melvin, "The Split Incentives Energy Efficiency Problem: Evidence of Underinvestment by Landlords," *Energy Policy, 115* (2018): 342, https://doi.org/10.1016/j.enpol.2017.11.069.
[6] Erica Myers, "Asymmetric Information in Residential Rental Markets: Implications for the Energy Efficiency Gap," *Journal of Public Economic* 190 (2020), https://doi.org/10.1016/j.jpubeco.2020.104251.

consumption habits (for instance by turning down thermostats or closing windows), as they do not bear the costs of their inefficient energy consumption.[7] In this scenario, landlords will invest in energy-efficient appliances to reduce the cost of renters' energy consumption, favouring long-term cost-savings over the initial cost of appliances.[8]

Split incentives between landlords and renters are driven by two key causes: asymmetrical information and inattentiveness. A prospective renter seeking to minimize cost will search for rental properties with higher energy efficiency, so long as they expect marginal energy savings to outweigh the increased costs of rent. However, asymmetrical information may lead to suboptimal energy efficiency choices as energy-related technologies in the unit are unobservable to the prospective renter,[9] or these differences are costly to observe.[10] Where renters pay utilities, renters may be unsure of the expected utility costs attached to the rental unit.[11] Even where the information on energy consumption may be available to a prospective renter or landlord, other facets of their decision-making process may be more salient at the time of choice than energy-efficient consumption.[12]

1.2 Financial Barriers to Renters

Renters account for a large and diverse population of residents in the City of Vancouver. According to the 2016 Census, renters represented more than half of all Vancouver households (53%) – a total of 150,745 households in Vancouver were renting while 133,160 households owned their homes.[13] Across household and family types, owner households tend to earn a higher income than renter households. While all income levels have been found in

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[7] Kenneth Gillingham, Matthew Harding and David Rapson, "Split Incentives in Residential Energy Consumption." *The Energy Journal* 33, no. 2 (2012): 37-62. http://www.jstor.org/stable/23268077.
[8] Ibid.
[9] James Carroll et al., "Low Energy Efficiency in Rental Properties: Asymmetric Information or Low Willingness-to-Pay?" *Energy Policy* 96 (2016): 617-629, https://doi.org/10.1016/j.enpol.2016.06.019.
[10] Hunt Alcott and Michael Greenstone, "Is There an Energy Efficiency Gap?" *Journal of Economic Perspectives* 26 no. 1 (2012): 3-28, https://doi.org/10.1257/jep.26.1.3.
[11] Maruejols and Young, "Split Incentives and Energy Efficiency in Canadian Multi-Family Dwellings," 3660.
[12] Alcott and Greenstone, "Is There an Energy Efficiency Gap?" 20.
[13] "City of Vancouver Housing Needs Report," City of Vancouver, Updated April 2022. https://vancouver.ca/files/cov/pds-housing-policy-housing-needs-report.pdf.

both categories, the median household income was $88,431 for owners compared to $50,250 for renter households.[14] Rental costs in Vancouver increased by over 45% between 2011 and 2021, and over 50% across the region.[15] A key factor in explaining income differences between owners and renters is variation in household composition. Census data indicates that owner households are more likely to be couples, often with a dual income, with or without children; a greater proportion of renter households are singles, followed by couples with no children.[16] Differences in household composition can greatly impact income, purchasing power, and affordability.[17] For this reason, it is important to consider the household type in our income and affordability analyses. For more information on household types, see Appendix A.

Because all income levels are found amongst renters, this will inevitably impact what climate reduction actions they can and will take. Their rental housing situation will also impact this–renters may live in condos, basement suites, laneway homes or multi-unit apartment buildings, among others.[18] In particular, barriers for renters, especially low-income renters, are evident when current incentive programs (such as heat pump rebates) benefit homeowners, but not renters.

While not all renter households are low-income, low-income households are disproportionally renters. Groups including Indigenous and racialized households, renting seniors, lone-parent households and people with accessibility needs are disproportionally low-income households.[19] Indigenous households also face "unique housing challenges

[14] "City of Vancouver Housing Needs Report," City of Vancouver, updated April 2022. https://vancouver.ca/files/cov/pds-housing-policy-housing-needs-report.pdf.
[15] Ibid.
[16] Ibid.
[17] Ibid.
[18] "Housing Characteristics Fact Sheet," City of Vancouver, updated April 30, 2017. https://vancouver.ca/files/cov/housing-characteristics-fact-sheet.pdf.
[19] Residents who are Indigenous and Black; seniors; lone-parent households; and 2S/LGBTQIA+ also disproportionately face housing insecurity or homelessness.

rooted in the legacy of colonialism."[20] While cutting carbon emissions across the City will require significant financial investment from various actors, a critical issue is that, in many cases, if renters are expected to reduce their household carbon emissions on their own accord, there are inevitable financial barriers that restrict their participation.

1.3 Insufficient Actor Engagement

Reducing renters' carbon emissions in the City of Vancouver poses another consideration in defining the policy problem: insufficient engagement with existing carbon reduction strategies. "Information asymmetry" arises when markets and governments insufficiently convey the "value of future energy savings" to renters and landlords in their undertaking of carbon emission reduction strategies.[21] Renters are unable to fully engage in carbon reduction strategies when they do not know what those strategies are in the first place, or what role they have in initiating them within landlord-tenant relationships.

The existence of *periodic* or *short fixed-term* tenancy agreements means that renters might lease a property for an insufficient duration to undertake some carbon emission reduction strategies. British Columbia tenancy law defines periodic tenancy agreements as having no specific end date, where either the landlord or renter may terminate the agreement (i.e., "month-to-month" tenancy), whereas a short fixed-term agreement leases a unit for a specific duration (i.e., "a year, month or a week").[22] Shorter tenancies may lead some renters to view undertaking even those permanent carbon emission reduction strategies within their control as exceeding the scope of their tenancy obligations, which threatens strategy engagement.[23] A

[20] "City of Vancouver Housing Needs Report."
[21] Ghislaine Lang and Bruno Lanz, "Energy Efficiency, Information, and the Acceptability of Rent Increases: A Multiple Price List Experiment with Tenants" (Working Paper, Massachusetts Institute of Technology Center for Energy and Environmental Policy Research, 2020), 1, https://ceepr.mit.edu/workingpaper/energy-efficiency-information-and-the-acceptability-of-rent-increases-a-multiple-price-list-experiment-with-tenants/.
[22] "Tenancy Agreements," Government of British Columbia, updated February 23, 2021, https://www2.gov.bc.ca/gov/content/housing-tenancy/residential-tenancies/starting-a-tenancy/tenancy-agreements.
[23] Ghislain Dubois et al., "It starts at home? Climate policies targeting household consumption and behavioral decisions are key to low-carbon futures," *Energy Research & Social Science* 52 (2019). https://doi.org/10.1016/j.erss.2019.02.001.

further problem contributing to insufficient landlord engagement is restrictive building regulations, especially for regulated condos, strata units, townhouses, rental apartments, or co-op buildings.[24] Rules governing retrofit procedures, renovations, or major building infrastructural changes currently diminish landlords' ability to undertake carbon emission reduction strategies.[25]

2. CONNECTING THE PROBLEM TO ACTION: GOALS

2.1 Primary Goal: Increase Renters' Agency to Engage in Carbon Emissions Reduction Strategies in the City of Vancouver

Decision-making agency will be measured by identifying *who* (renters, landowners) can engage with carbon reduction strategies and *what* level of information is available to the decision-maker. Renters' agency is considered strengthened when renters can make their own decisions on carbon reduction strategies, are accurately informed on the available carbon reduction strategies, and have increased participation in carbon reduction strategies.

2.2 Secondary Goal: Reduce renters' carbon emissions at the most cost-effective rate

This goal aims to ensure that policy alternatives maximally reduce carbon emissions in relation to their costs for landlords, renters, and governments.[26] Broadly framing cost-effectiveness expands what shape policy alternatives might take. Assessing how well policy alternatives satisfy this objective might incorporate some combination of two cost-effectiveness tests: (1) a Program Administrator Cost (PAC) test, and (2) a Societal Cost Test (SCT).

The PAC test assesses carbon reduction strategies' cost-effectiveness by permitting strategy funders (i.e., landlords, renters, governments) to evaluate the strength of strategy

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[24] "Energy resources and programs for existing multi-family buildings," City of Vancouver, updated April 26, 2022, https://vancouver.ca/green-vancouver/energy-resources-and-programs-for-multi-family-buildings.aspx.

[25] "Design Guidelines and Construction Standards," BC Housing, updated May 2019, https://www.bchousing.org/projects-partners.

[26] Melanie Gilarsky, Kriston McIntosh, and Jay Shambaugh, "How to reduce emissions as much as possible at the lowest cost," *The Brookings Institution*, April 21, 2020, https://www.brookings.edu/blog/up-front/2020/04/21/how-to-reduce-emissions-as-much-as-possible-at-the-lowest-cost/.

outcomes—like renters' overall carbon emissions reduction—against the expenditures to realize them.[27] This test is useful for persuading combinations of landlords, renters, or governments to financially invest in carbon emission reduction strategies by outlining anticipated fiscal benefits.

The SCT quantifies externality benefits that are not represented in market prices, as well as other non-carbon emission reduction benefits.[28] This test helps evaluate cost-effectiveness in relation to a wider network of social benefits, like energy savings, additional resource-saving, and non-carbon-reduction-related benefits while assessing reduction strategies' operational, installation, and overhead costs.[29]

Taken together, these tests qualify carbon emission reduction strategies differently for diverse actors. For example, PAC evaluations might better persuade market-motivated actors to undertake carbon emission reduction strategies, whereas SCT evaluations might better persuade those pursuing a broader, not wholly market-based agenda.

2.3 Secondary Goal: Equity as a Priority: Ensure All Renters Have Equitable Access to Carbon Reduction Strategies in The City of Vancouver

We cannot ignore that systemic discrimination and past policy decisions, including urban planning policies, have not sought to ameliorate inequality and have instead contributed to the ongoing oppression of Indigenous people, racialized, and other marginalized communities.[30] [31] As a result, certain communities in the City of Vancouver are more impacted by poverty, lack of services and unequal opportunities. These communities also will

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[27] Megan Billingsley et al., "The Program Administrator Cost of Saved Energy for Utility Customer-Funded Energy Efficiency Programs" (Technical Report, Ernest Orlando Lawrence Berkeley National Laboratory, 2014), 42, https://doi.org/10.2172/1129528.
[28] Martin Kushler, "Cost-Effectiveness Testing for Energy Efficiency Programs" (lecture, American Council for an Energy-Efficient Economy, Washington D.C., February 11, 2021). https://www.energy.gov/sites/default/files/2021/03/f83/bbrn-peer-test-021121.pdf.
[29] Ibid.
[30] "Council Report: Climate Emergency Action Plan."
[31] Fitzgerald, Joan. "Transitioning From Urban Climate Action to Climate Equity" *Journal of the American Planning Association* 88, no. 4 (2022): 508-523. https://doi.org/10.1080/01944363.2021.2013301.

be more impacted by the effects of climate change. For instance, in late June 2021, when British Columbia experienced an unprecedented extreme heat event, nearly 30% of those who died lived in materially and socially deprived neighbourhoods.[32] This also means, as the City strives to implement ambitious carbon reduction strategies, there will be residents who will experience barriers and unequal access to contribute their share.[33] Carbon reduction policies may even disadvantage or under-serve communities with limited access to lower carbon options.[34] Simply, variations in income amongst residents in the City exist. This reflects a need for tailored approaches to reducing renters' carbon emissions. We know that some renters, with proper motivation and empowerment and reason to, do have the means to reduce their carbon emissions in their households through a variety of actions. However, we also know that lower-income renters are less likely to be in that position. This presents a barrier to access we must consider. While low-income renters may want to be part of the solution, their participation may be constrained by unequal access to lower-carbon options. Our goal is to ensure all renters have equitable access to carbon reduction strategies in the City. As we work towards this objective, we will place equity as a priority.

While it is difficult to measure if all renters have access, one way to measure this objective would involve a period of engagement, where feedback could be collected, and a survey or questionnaire could be conducted. This would involve surveying renter households to determine if they have engaged in carbon reduction strategies and how high of a priority these strategies are to them. The survey would be accessible online and available in multiple languages. A key equity consideration is dialogue with those who are more likely to

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[32] Government of British Columbia. "Extreme Heat and Human Mortality: A Review of Heat Related Deaths in B.C. in Summer 2021." Report to the Chief Coroner of British Columbia. June 7, 2022. https://www2.gov.bc.ca/assets/gov/birth-adoption-death-marriage-and-divorce/deaths/coroners-service/death-review-panel/extreme_heat_death_review_panel_report.pdf.

[33] Friend, Richard and Marcus Moench. "What is the purpose of urban climate resilience? Implications for addressing poverty and vulnerability." *Urban Climate Volume* 6 (2013): 98-113. https://doi.org/10.1016/j.uclim.2013.09.002.

[34] Jacqueline Wilson, "Ontario's Climate Change Plan: Protecting the Environment and Low-Income Communities," *Canadian Environmental Law Association*, Updated November 16, 2018.

experience barriers to accessing lower carbon options. In our preliminary analysis, we will pay close attention to potential impacts on lower-income renters, but further dialogue will be needed with different renter sub-populations to adjust and implement policy alternatives.

3. REDUCING RENTERS' CARBON FOOTPRINT IN THE CITY OF VANCOUVER

Reducing renters' carbon footprint is possible to the extent that they are willing and able to pay through different reduction strategies. Tracking the reduction of GHG emissions of policy alternatives is important to reach the City of Vancouver's net-zero goals. To measure the reduction of renters' carbon footprint in the City of Vancouver, this memo calculates the increase in fossil fuel consumption caused by the issue of split incentives.[35] We estimate the total units of fuel currently being consumed from renters' activity, multiplied by the associated carbon emissions. Then, we subtract the projected fuel consumption, multiplied by the associated carbon emissions to determine the total reduction of emissions.

POLICY ALTERNATIVES

We evaluate each policy alternative and the consequences of not taking action in Appendix D according to our four goals and the measurements we propose in the preceding section.

Rental Units Carbon Footprint Report Card

British Columbian renters are at present encouraged to find out utility costs before renting a home, as utility bills can add hundreds of dollars per month in addition to rent.[36] However, no policy at the City or provincial level requires disclosing this information before entering into rental agreements. This restricts available information to renters on actual rental unit costs, especially when utility costs are not the foremost concern when choosing rental units.[37] We suggest the implementation of a Rental Units Carbon Footprint Report Card to empower

[35] Jesse Melvin, "The Split Incentives Energy Efficiency Problem: Evidence of Underinvestment by Landlords."
[36] "Paying Rent," Tenant Resource and Advisory Centre," accessed November 13, 2022. https://tenants.bc.ca/your-tenancy/paying-rent/#the-basics.
[37] Alcott and Greenstone, "Is There an Energy Efficiency Gap?."

renters by informing them of energy efficiency, utility costs and carbon emissions associated with prospective rental units. This would empower renters to make informed choices when choosing rental units based on associated utility costs and their carbon outputs. It is also expected to encourage at least a subset of landlords to invest in energy-efficient systems to compete on the rental market.

A Rental Unit Carbon Emissions Footprint Report Card would include a ranking on the efficiency of energy usage in a unit, as well as the carbon dioxide emissions output. It should be made available before or during a rental unit viewing, provided again before any tenancy agreements are signed, and include the average carbon consumption needed to reach Vancouver's 2030 emissions targets, as the 'warm glow' effect should increase the motivation to pursue a carbon-friendly unit.[38] Appendix B outlines the calculation methods for rental unit energy efficiency and carbon footprint for this report card. The requirements and calculation methods are based on the Building Energy Rating Certificate, a similar program implemented in Ireland, required for all rental units. This may require adjustment for the BC context.[39] This report card would be required prior to any tenancy agreements is signed, and would be encouraged when the unit is listed on the market.. Adjustments to the Residential Tenancy Act fall under provincial jurisdiction.[40] The City could push for provincial action on this issue, to protect renters within Vancouver.

Renter Services Centre in the City of Vancouver

The City of Vancouver is proposing to create a community-based Renter Services Centre (RSC). The proposed multi-service centre would provide information on renters' rights; legal advice and advocacy; emergency financial assistance; and assistance navigating basic renter

[38] James Carroll et al., "Low Energy Efficiency in Rental Properties: Asymmetric Information or Low Willingness-to-Pay?" *Energy Policy* 96 (2016): 617-629, https://doi.org/10.1016/j.enpol.2016.06.019.
[39] "Understand a BER," Sustainable Energy Authority of Ireland, accessed November 13, 2022, https://www.seai.ie/home-energy/building-energy-rating-ber/understand-a-ber-rating/.
[40] Residential Tenancy Act, 2002. https://www.bclaws.gov.bc.ca/civix/document/id/complete/statreg/02078_01#section12.

issues, including maintenance and repair.[41] Should the City approve the implementation of this centre in the Fall 2022, we propose that rental unit carbon emission reduction information should be available at this centre alongside other information services. The basic climate action and green buildings information RSC would provide, communicated via a staff member or a printed-out pamphlet:

Immediate carbon emission reduction strategies (i.e. lower the thermostat, ensure the heating system in the unit is working efficiently)	*Investing in future carbon emission reduction strategies* (i.e. switching to electric appliances, like heat pumps or induction stovetop)

No similar information and referral hub currently exists in the City of Vancouver.[42] In relation to city-wide climate action, information for renters is essential. This is particularly crucial as a central hub for lower-income renters to access information when other means are not available to them. The need for information-sharing from a trusted person or organization in the community is necessary to achieve climate justice.[43] A place-based approach to accessing this information will increase renters' knowledge of the threat posed by climate change, which in turn will increase their agency. The City of Vancouver is actively exploring the value and feasibility of creating an RSC.[44] Thus, the political and operational practicality of this policy alternative depends on the outcome of the RSC proposal. However, providing renters with relevant carbon reduction strategies information in their rental units could be provided at alternative locations, such as public libraries and community recreation centres.

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[41] "Council Memo - Renter Services Centre Update," City of Vancouver, updated April 25, 2022, https://vancouver.ca/files/cov/2022-04-26-council%20memo-renter-services-centre-update-rts-14762.pdf?_ga=2.29775162.487200905.1667943705-1147175993.1667777439.

[42] Ibid.

[43] Agyeman, Julian, Bob Doppelt, Kathy Lynn, and Halida Hatic. "The Climate-Justice Link: Communicating Risk with Low Income and Minority Audiences," in *Creating a Climate for Change: Communicating Climate Change and Facilitating Social Change*, ed. Susanne C. Moser and Lisa Dilling (Cambridge: Cambridge University Press, 2007), 119–38. https://doi.org/10.1017/CBO9780511535871.

[44] "Council Memo - Renter Services Centre Update," City of Vancouver, updated April 25, 2022, https://vancouver.ca/files/cov/2022-04-26-council%20memo-renter-services-centre-update-rts-14762.pdf?_ga=2.29775162.487200905.1667943705-1147175993.1667777439.

A Small Appliances Loan Program in the City of Vancouver

An accessible loan program would allow renters to borrow and test portable, carbon-efficient appliances. This includes induction cooktop stove: a safer, more energy-efficient option for everyday stovetop cooking.[45] This lending program would operate out of one of the City's community centres or libraries, potentially at multiple locations. This ensures the program is accessible by all transportation methods. This equipment-sharing allows renters to try out energy-efficient small appliances at no cost to them for a fixed, renewable borrowing duration. It would also allow them to return the appliance should their tenancy end. Appliances may then be re-used, rather than disposed of. We have seen this lending concept implemented elsewhere. The City of San Jose, California has introduced an Induction Cooktop Checkout program, where residents can check out a portable induction cooktop stove for up to two weeks at no cost.[46] We too have seen a project underway in the City of Vancouver–*The Thingery*–a community-supported equipment-lending library.[47] Based on other models, the general goal is to implement an appliance-sharing program in the City. This would take allotting a fixed amount from other City revenue reserves to fund the development of this program, including the purchase of induction stoves for rentals and additional staffing needs, and further development of where this program would operate.

A Small Investment Grant for Renters

This policy alternative would create a small investment fund to provide renters with microgrants to immediately modify carbon-inefficient home features or reduce carbon emissions by purchasing carbon-efficient equipment. Carbon emission reduction strategies

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[45] O'Shea-Evans, Kathryn. "Why New Induction Cooktops are Safer and Faster than Gas Or Electric; rather than Relying on Flames Or Piping Hot Burners, these High-Tech Ranges use Electromagnetism to Heat the Bottom of Pans Directly. here, the Pros and Cons." *Wall Street Journal (Online),* Apr 22, 2021. https://www.proquest.com/newspapers/why-new-induction-cooktops-are-safer-faster-than/docview/2516253626/se-2.

[46] "Induction Cooktop Checkout Program," City of San Jose. Accessed November 7, 2022. https://www.sanjoseca.gov/your-government/departments-offices/environmental-services/climate-smart-san-jos/induction-cooktop-checkout-program.

[47] "The Thingery," The Thingery. Accessed November 8, 2022. https://thethingery.com/.

include (but are not limited to) purchasing induction hot plates for gas-free cooking,[48] installing thermal curtains or window inserts to prevent heat from escaping through windows,[49] or replacing halogen lightbulbs with carbon-efficient LEDs.[50] This investment fund borrows from similar funding arrangements between renters and the provincial government–like the Rental Assistance Program–to incorporate income-based criteria for microgrant qualification.[51] Considering renters' income level in allocating investment funds focuses financial support to renters with limited economic resources to undertake carbon emission reduction strategies as opposed to wealthier renters who could do so anyways.

Targeting immediate carbon emission reduction strategies simultaneously benefit renters, landlords, and city officials. Renters who qualify for microgrants can update carbon-emitting equipment for limited or no out-of-pocket expenses. Landlords who pay utility costs may experience lower monthly payments due to more efficient renter practices and equipment usage. City officials might gain political points from renters and landlords for these above benefits while working toward reducing the City of Vancouver's total carbon emissions.

Revised Standards of Maintenance Bylaw to Mandate Carbon-Reducing Appliances

This policy alternative adds regulatory subclauses to the City of Vancouver Standards of Maintenance Bylaw that would mandate replacing high carbon-emitting appliances with approved carbon-efficient alternatives once they fail.[52] Landlords will eventually replace appliances given their mechanical lifespan and revising this bylaw capitalizes on a policy window to make landlords select approved carbon-efficient replacements. An approved

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[48] Hamed M. El-Mashad and Zhongli Pan, "Application of Induction Heating in Food Processing and Cooking," *Food Engineering Reviews* 9 (2017): 87-88. https://doi.org/10.1007/s12393-016-9156-0.

[49] Daniel Sean Mistro, "Window Inserts and the People Adopting them: Building Sustainable Communities in Maine" (MSc thesis, University of Maine, 2017), ProQuest Dissertations and Theses Global (10821176).

[50] Oluwatobi G Adekanye, Alex Davis, and Inês L Azevedo, "Do LED lightbulbs save natural gas? Interpreting simultaneous cross-energy program impacts using electricity and natural gas billing data," *Environmental Research Communications* 3, no. 1 (2021): 12. https://doi.org/10.1088/2515-7620/abddc7.

[51] "Rental Assistance Program (RAP)," BC Housing, updated July 26, 2022, https://www.bchousing.org/housing-assistance/rental-assistance-programs/RAP.

[52] The departments of (a) *Planning, Urban Design, and Sustainability*, (b) *Development, Buildings and Licensing*, and (c) *Engineering Services* might work in unison to approve a carbon-efficient appliance catalogue in the Office of the City Manager.

catalogue can prioritize non-gas-powered appliance options, reducing total building or unit carbon emissions at a relatively unintrusive rate to match landlords' inevitable actions.

This policy alternative would add a subclause to § IV, *Maintenance of Land, Buildings, and Accessory Building*s, which mandates that landlords regularly monitor and replace failing appliances in any building or unit.[53] The new subclause would outline landlords' obligations to replace the defunct appliances, further detailing the Government of British Columbia's support for recycling large appliances under the *Environmental Management Act*.[54] An additional subclause to § XVII, *Gas Appliances and Systems*, specifies conditions for maintaining and replacing ***gas*** appliances,[55] centring replacement on the highest carbon-emitting appliances.[56] See Appendix C for potential subclause language.

NEXT STEPS

Recommendation

We recommend implementing a "Rental Unit Carbon Emissions Report Card" program, as it is the most *feasible* and *actionable* of the policy alternatives presented to reduce renters' carbon footprints in the City of Vancouver. We determined the feasibility and actionability of this recommendation with consideration to our four policy goals and their associated measurements (see Appendix D for detailed policy alternative evaluations). A "Rental Unit Carbon Emissions Report Card" program delivers a high degree of renter agency; high program administrator cost effectiveness; comparatively high societal cost effectiveness; high equity across renter stratifications (i.e., age, income levels, rental unit types, etc.); and comparatively high ability to reduce renters' carbon footprints.

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[53] City of Vancouver Standards of Maintenance Bylaw, 2022, no. 5462, c. 4, https://app.vancouver.ca/bylaw_net/ConsolidatedReport.aspx?bylawid=5462&txtSearch=5462.
[54] Environmental Management Act, S.B.C. 2003, § III, s. 25, https://www.bclaws.gov.bc.ca/civix/document/id/complete/statreg/03053_03#section25.
[55] City of Vancouver Standards of Maintenance Bylaw, 2022, no. 5462, c. 17, https://app.vancouver.ca/bylaw_net/ConsolidatedReport.aspx?bylawid=5462&txtSearch=5462.
[56] Statistics Canada, *Household food consumption and Canadian greenhouse gas emissions, 2015*, by Jennie Wang and Abdoul-Razak Mamane, Catalogue no. 16-508-X (Ottawa, ON: Statistics Canada, 2019), https://www150.statcan.gc.ca/n1/pub/16-508-x/16-508-x2019004-eng.htm.

The "Rental Unit Carbon Emissions Report Card" would be mandated to require landlords to share information on utility costs and carbon outputs with prospective renters. This solution would enable renters to make more informed choices when choosing rental units, based on associated utility costs and their carbon outputs, while potentially encouraging landlords to invest in energy-efficient systems to compete in a more transparent rental market. This recommendation also considers the City of Vancouver's pro-development and entrepreneurial history by mapping the "Rental Unit Carbon Emissions Report Card" onto the existing market activity of the region. Given that this recommendation simply requires landlords to disclose carbon emissions-related information to increase transparency, the "Rental Unit Carbon Emissions Report Card" program is a politically attractive policy alternative to support the City of Vancouver's broader efforts to reduce carbon emissions.

Preliminary Advice on an Implementation Strategy

We advise that landlords be required to provide a Report Card to prospective renters. Some Report Card features would include:

1. **Energy Grade**: a letter assigned based on kilowatt hours per square metres (kWh/m2)
2. **Energy Efficiency:** the estimated energy use per unit area, per year
3. **Carbon Dioxide Emissions:** the associated carbon dioxide output, related to energy use and energy sources (gas or electric) within a unit.

Renters may live in condos, basement suites, laneway homes or multi-unit apartment buildings and other housing arrangements.[57] This report card system will be individually reported by homeowners and landlords of laneway homes and basement suites, or commercially reported by organizations in the case of multi-unit buildings and developments. Implementing a report card system will require a fixed implementation cost and subsequent auditing costs. The upcoming Capital Plan contemplates $460 million of capital investment to advance the Climate Emergency Action Plan and the accompanying Climate Change

[57] "Housing Characteristics Fact Sheet," City of Vancouver, updated April 30, 2017. https://vancouver.ca/files/cov/housing-characteristics-fact-sheet.pdf.

Adaptation Strategy, and $260 million more to deliver additional climate mitigation and adaptation benefits.[58] To this, the City of Vancouver's 2023 – 2026 Capital Plan states climate mitigation and adaptation work as a key priority.[59] As a climate action item, we recommend that the implementation costs should be allocated from the approved climate action budget.

For initial implementation advice, we may look to the Building Energy Rating Certificate, a successful program implemented in Ireland.[60] This program required all rental units in Ireland to list the building energy rating. This would be adjusted to the BC and City of Vancouver context. By mandating that BERs be provided in rental unit listings, the Building Energy Rating (BER) program in Ireland saw the probability of a rental unit listing their BER certificate increase by 62.8%.[61] This is a low uptake despite an associated penalty of up to £5 000 ($9 000 CAD) for non-compliance. The City of Vancouver should consider raising the penalty for non-compliance, and increasing awareness amongst landlords and renters of the report card requirement, when implementing the current proposal.

Currently British Columbia's Ministry of Energy, Mines and Low Carbon Innovation is discussing the idea of a "Home Energy Labels" for homeowners. This also presents an opportunity to implement our report card (energy labels) to get this information to renters that could be explored.

Projected Outcomes

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58 Ibid.

59 "Special Council – 2023-2026 Capital Plan: Final Plan," City of Vancouver, June 2022. https://council.vancouver.ca/20220629/documents/spec1.pdf?_ga=2.83334303.825834422.1659482212-647602582.1659482212.

60 "Understand a BER," Sustainable Energy Authority of Ireland, accessed November 13, 2022, https://www.seai.ie/home-energy/building-energy-rating-ber/understand-a-ber-rating/.

61 Marie Hyland, Ronan C. Lyons and Sean Lyons, "The Value of Domestic Building Energy Efficiency - Evidence from Ireland," *Energy Economics* 40 (2013): 943-952. https://doi.org/10.1016/j.eneco.2013.07.020

We predict that the Renters' Report Carbon would enable renters to make more informed choices when choosing rental units, based on associated utility costs and their carbon outputs, while potentially incentivizing landlords to invest in energy-efficient systems to compete in a more transparent rental market. It is important, however, to note the effects of an energy ranking system on housing affordability: studies conducted within Ireland have demonstrated that energy efficiency ratings have an upwards effect on the rental price of energy-efficient units. In Ireland, an A-Rated property experiences rental rates that are 1.8% higher than D-Rated properties,[62] suggesting that Vancouver should expect to see higher costs for more energy-efficient units. While about half of Irish renters are willing to pay more for highly energy efficient properties, those who were unwilling to pay cited inability to afford higher rents as their main reason.[63] In Vancouver, we should expect higher-income renters to make choices based off of the Report Card, however we expect to see more limited effects on choice for lower-income renters. Lower-income renters are still empowered to make choices regarding utility costs of prospective units, especially as gas prices rise. The City should continue to explore options to empower low-income renters to decrease their carbon footprint, alongside the Report Card.

Collecting rental unit carbon emissions data through a landlord-renter reporting program also delivers new insights into carbon emissions distribution and magnitude across the City. This data will be useful for future carbon emission reduction initiatives, or suggest spatial areas for targeted policy or initiative focus. Collecting this data is also a first step to implementing mandatory minimum energy and carbon efficiency requirements for all units in the City.

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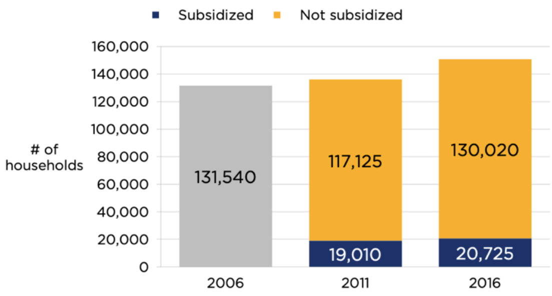
[62] Hyland et. al, 2013.
[63] Matthew Collins and John Curtis, "Return on Energy Efficiency Investments in Rental Properties," *ESRI Research Bulletin* (2018): 1-4. https://aei.pitt.edu/101917/1/RB201816.pdf

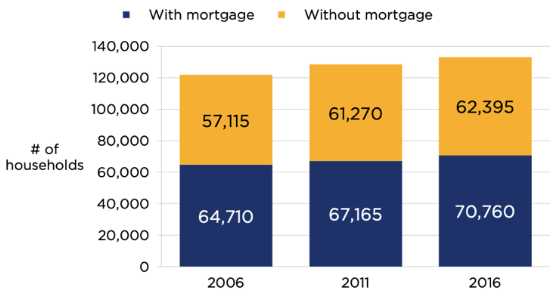
Ultimately, this Report Card is one step towards the carbon reductions needed by buildings in the City of Vancouver to reach net-zero by 2030, and positioning the City of Vancouver as a global climate leader.

Appendix A

Subsidized and non-subsidized renter households in Vancouver, 2006 – 2016



Presence for mortgage for owner households in Vancouver, 2006-2016



Source: National Household Survey, 2016, Statistics Canada, Ottawa ON. Retrieved from Housing Vancouver Strategy, 2019, City of Vancouver. https://vancouver.ca/files/cov/2019-housing-vancouver-annual-progress-report-and-data-book

Appendix B

Rating Section	**Calculation Method**
Energy Grade	Scale ranging from A+ to F, where: 1. A+ is equal to or under 25 kWh/m^2 2. A is under 50 kWh/m^2 3. A- is under 75 kWh/m^2 4. B+ is under 100 kWh/m^2 5. B is under 125 kWh/m^2 6. B- is under 150 kWh/m^2 7. C+ is under 175 kWh/m^2 8. C is under 200 kWh/m^2 9. C- is under 250 kWh/m^2 10. D+ is under 275 kWh/m^2 11. D is under 300 kWh/m^2 12. D- is under 350 kWh/m^2 13. F is over 400 kWh/m^2
Energy Efficiency	Estimated energy use per unit area, per year, based upon: 1. Set heating pattern across Vancouver 2. Estimated number of occupants within a unit 3. Building construction type 4. Year of construction 5. Floor dimensions 6. Room height 7. Window and door sizes 8. Level of insulation in wall, roof, and floor 9. Space heating system and heating controls 10. Hot water systems and water cylinders 11. Ventilation characteristics 12. Baths and showers 13. Lighting fixtures 14. Renewable systems within units.
Carbon Dioxide Emissions	Associated carbon dioxide output, related to energy use and energy sources (gas or electric) within a unit. Included with the necessary carbon reductions needed by buildings in the city of Vancouver to reach net-zero by 2030.

Appendix C

Subclause 4.1.14 - Maintenance of Land, Buildings, and Accessory Buildings:

4 (...)

(14) Reaching expiry or failing, building appliances–including but not limited to refrigerators, stoves, ovens, dishwashers, clothes washers, and clothes dryers–must be replaced with a similar carbon-efficient appliance approved by the City of Vancouver.

Defunct building appliances are to be recycled free of charge under the Province of British Columbia's Large Appliance Recycling Program.

Subclause 17.1.4 - Gas Appliances and Systems:

17 (...)

(4) Reaching expiry or failing, gas appliances and systems–including but not limited to stoves, ovens, cooktops, ranges, refrigerators and freezers, and space heaters–are to be replaced with a similar carbon-efficient appliance or system approved by the City of Vancouver.

Appendix D

Evaluation of Policy Alternatives

POLICY GOALS

	Increase Renters' Agency	**Cost-Effectiveness**	**Equitable Access to Carbon Reduction Strategies**	**Renter's Carbon Footprint Reduction**
Rental Units Carbon Footprint Report Card	This policy alternative scores **HIGH** because renters *will be accurately informed* on the carbon output and energy efficiency of rental units, and *be empowered* to make informed choices when choosing rental units.	This policy alternative scores **HIGH** in projected program administrator cost (PAC), given its comparatively low implementation costs yet persisting administration and monitoring (auditing) costs. This policy alternative scores **MEDIUM** in the projected societal cost test (SCT), given the potential delays arising from the landlords' slower response to replace high carbon emitting appliances to compete in rental markets.	This policy alternative scores **HIGH** because every prospective renter will receive this information. Based on our measurement, a period of engagement may also be necessary to encourage dialogue and to adjust the Report Card to meet low-income renters' needs.	This policy alternative scores **MEDIUM** because while it does increase the availability of energy efficiency information, and may increase uptake in energy-efficient technology in rental units, it is dependent on the *willingness to pay* of landlords and renters to adopt energy-efficient technology.
Renter Services Centre	This policy alternative scores **MEDIUM** because renters will be *accurately informed* on the carbon reduction strategies, but not necessarily be *empowered* to undertake these alternatives.	This policy alternative scores **MEDIUM** in projected program administrator cost (PAC), given its development costs, persisting administration, and service delivery costs. This policy alternative scores **LOW** in the projected societal cost test (SCT), given that this	policy alternative scores **MEDIUM** because although all renters may have access to carbon-reduction information, the Centre is a place-based alternative that might not necessarily be accessible to all. Renters may also not be able to afford to adopt strategies that are recommended to them at the Centre.	This policy alternative scores **LOW** because while it does increases the availability of information on carbon reduction strategies, reducing carbon emissions depends on renters' willingness to visit the centre and to adopt recommended strategies. It is difficult to guarantee that any action will be taken to reduce

POLICY GOALS				
		provision of carbon-reduction *information* does not necessarily lead to *actual* carbon reduction.		renters' carbon footprint.
Small Appliances Loan Program	This policy alternative scores **MEDIUM** because renters will be *empowered* to undertake small appliance upgrades, but not necessarily have the agency to change 'big-ticket' appliances (i.e. furnaces) within rental units. Renters may also not be *informed* of carbon reduction alternatives.	This policy alternative scores **MEDIUM** in projected program administrator cost (PAC), given its program implementation costs (i.e., purchasing small appliances to loan) and persisting administration and monitoring costs. However, the City of Vancouver retains assets in the ownership of carbon-efficient appliances. This policy alternative scores **HIGH** in the projected societal cost test (SCT), given the immediate carbon emission reductions associated with borrowing and using efficient appliances.	This policy alternative scores **MEDIUM** because the loan program is placed-based, which might not necessarily be accessible to all. As well, it requires time to seek out, and then return the small appliances.	This policy alternative scores **MEDIUM** because while it decreases the cost barrier to carbon-friendly appliances to renters, it does not target 'big-ticket' infrastructure items producing carbon emissions in buildings (i.e. gas furnaces), and does not guarantee uptake in carbon-friendly small appliances.
Small Investment Grant for Renters	This policy alternative scores **MEDIUM** because renters will be *empowered* to undertake small unit upgrades, but not necessarily have the agency to change 'big-ticket' appliances (i.e. gas furnaces) within rental units. Renters may also not be *informed* of carbon reduction alternatives.	This policy alternative scores **MEDIUM** in projected program administrator cost (PAC), given the high costs of providing targeted funds to qualifying renters, however, efficiency in mediating renters' carbon emission reduction strategies. This policy alternative scores **HIGH** in the projected societal	This policy alternative scores **HIGH** because all renters may apply and there are barriers to receiving this grant.	This policy alternative scores **MEDIUM** because while it decreases the cost-barrier to carbon-friendly energy upgrades to renters, it does not allow renters to target 'big-ticket' infrastructure items producing carbon emissions in buildings (i.e. gas furnaces), and does not guarantee uptake in carbon-friendly upgrades by

POLICY GOALS					
			cost test (SCT), given that it affirmatively enables renters to undertake carbon emission reduction strategies.		renters.
	Revised Standards of Maintenance Bylaw	This policy alternative scores **LOW** because renters are not *informed* on different strategies of carbon reduction, and are not *empowered* to undertake these strategies. The obligation is placed on the landlord.	This policy alternative scores **HIGH** in projected program administrator cost (PAC), given its This policy alternative scores **MEDIUM** in the projected societal cost test (SCT), given that mandating the replacement of carbon-inefficient appliances will affirmatively reduce total carbon emissions but at some costs to landlords.	This policy alternative scores **LOW/MEDIUM** because it puts the onus solely on landlords to undertake carbon reduction strategies.	This policy alternative is **HIGH** as it regulates the number of carbon emissions rental units is legally allowed to produce in the City of Vancouver.
	No Action	This policy alternative scores **LOW** because renters are not currently *informed* on different strategies of carbon reduction, and are not *empowered* to undertake these strategies.	This policy alternative scores an **INCONCLUSIVE** projected program administrator cost (PAC), considering that undertaking no action produces no actual costs to administrators. This policy alternative scores **LOW** in the projected societal cost test (SCT), given that not undertaking carbon emission reduction actions is unlikely to counter rising carbon emissions.	This policy alternative **LOW** because inequalities currently exist in certain communities are currently impacted by poverty, lack of services and unequal opportunities	This policy alternative **LOW** because no concrete actions will be implemented to reduce renters' carbon footprint. The current 53% of residents who are renters will continue contributing to the approximately 54% of emissions that derive from buildings will remain the same or even increase.

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Google Form Data

2. If yes, what is your improvement plan and availability? Please explain your plan following the three general directions for improvement: (1) Content improvement: such as adding more research, interviews, surveys, curation of useful resources, etc.; (2) Quantitative improvement: such as getting more data mining, database construction, data analysis, and data visualization effort; (3) Social impact and awareness improvement: working with GLOCAL social media specialists to boost your social media knowledge mobilization and awareness campaigns through creative content generation and social media ads.

1. Content Improvement: We will enhance the project's depth and relevance by incorporating additional research, conducting more interviews, and deploying surveys with renters, landlords, policymakers, and local government officials in the City of Vancouver. This will be done to gather comprehensive insights and experiences from the community that will inform an updated rubric for our proposed Carbon Emissions Report Card.

2. Quantitative Improvement: Our quantitative improvement approach involves enhancing our database construction efforts, improving data analysis, and augmenting data visualization techniques. This process includes a detailed examination of Vancouver's unique neighbourhood and spatial aspects to select ideal locations for trialing policy pilot projects. Additionally, we will focus on pinpointing specific demographic groups for in-depth interviews and surveys. This comprehensive approach aims to yield a more nuanced and detailed understanding of the effectiveness and impact of our strategies.

3. Social Impact and Awareness Improvement: Collaborating with GLOCAL social media specialists, we aim to amplify the project's reach and impact. This will be achieved by creating engaging content and utilizing targeted social media advertising across platforms like Instagram, LinkedIn, TikTok, Facebook, and YouTube. The goal is to enhance awareness and mobilize knowledge on the effects of and mitigation strategies for reducing renters' carbon footprints in Vancouver, as well as other jurisdictions. Further, we aim to organize a panel event with a series of academic, policy, and community experts on reducing renters' carbon footprints in densely populated urban contexts, like Vancouver. The event will be hosted virtually to maximize participation, and include a presentation and question-and-answer period based on our project.

The project will culminate in the development of an interactive website, serving as a central hub for information dissemination, interaction, and community engagement. This digital platform will play a pivotal role in reaching a wider audience, facilitating easier access to information, and fostering a more informed and engaged community. We will work with the GLOCAL social media specialists to boost engagement with, and participation in, future GLOCAL initiatives.

3. Do you have any questions or challenges?

None